

IVAYLO VALIROV

PROCESS ENGINEERING | PROCESS DEVELOPMENT | PILOT PLANT | MANUFACTURING

Fremont, CA | (858) 610-4937 | ivaylo.valirov@gmail.com | www.ivaylo.com

PROFESSIONAL SUMMARY

Process engineer with over a decade of GMP experience spanning operations, engineering, project management, maintenance, bioprocess scale-up, facility and equipment commissioning, and manufacturing data systems. Combines strong analytical and documentation skills with hands-on technical expertise and a results-driven approach.

CORE COMPETENCIES & SKILLS

Process: Fermentation and cell culture, centrifugation, homogenization, purification, filtration, TFF, lyophilization, spray drying, process utilities (WFI, process gases, steam, chilled water)

Operations: Production scheduling, capacity planning, resource allocation, inventory management, equipment training, cross-functional coordination, CDMO oversight, mentoring, continuous improvement, Lean, Six Sigma, 5S, Kaizen, Kanban, value-stream mapping, visual management, RCA (5 Whys and Ishikawa analysis)

Engineering: PFDs, P&IDs, equipment sizing and selection, process optimization, design reviews, utility assessments, automation integration, hands-on troubleshooting, preventive maintenance and calibration, reliability

Development: Scale-up, parameter screening, DoE, design space, CPP/CQA identification, control strategy, SPC

Projects: Facility startup, equipment specification, procurement, FAT/SAT, commissioning, IQQ, PPQ, validation

Documentation & Compliance: SOPs, batch records, deviations and investigations, CAPA, change control, URS, FMEA, traceability matrices, 21 CFR Parts 11, 210, and 211, ICH Q7-Q12 guidelines, internal and FDA inspections

Computer: Excel, Python, Power BI, SQL, JMP, SAP, TrackWise, Veeva, Smartsheet, Claude Code, Codex

Automation: DeltaV, MES, Ignition SCADA, FactoryTalk, PI, PLCs, autosamplers, Python, web app development

PROFESSIONAL EXPERIENCE

Staff Engineer, Pilot Plant | Vaxcyte, San Carlos, CA

Mar 2024 - Present

Pilot Plant Operations

- Launched a new fermentation pilot facility, leading equipment specification, procurement, installation, and commissioning.
- Developed pilot USP documentation, including batch records, ELN templates, work instructions, and automation recipes, to support compliant, error-free execution.
- Trained staff on equipment operation and routine maintenance.
- Designed and built a custom, open-architecture online data historian integrating seven standalone units from different manufacturers into one unified system, avoiding more than \$100K in licensing fees.
- Developed SQL- and Excel-based data systems and dashboards to standardize process development, pilot, and manufacturing data, reducing manual workflows by more than 90%.
- Mentored junior scientists and engineers, strengthening technical execution and team capability.
- Authored process development protocols and reports to support technology transfer to the manufacturing site.
- Developed internal systems for raw-material and freezer inventory management.
- Evaluated new PAT technologies (in-line cell density, autosamplers) to assist in process characterization and optimization.

Clinical Manufacturing USP MSAT

- Assumed an additional role as the USP MSAT representative for a CDMO clinical manufacturing facility, serving as the technical authority and decision-maker for process matters, document reviews, and quality records.
- Led technical USP meetings with the CDMO to address scale-up issues and yield optimization.
- Routinely presented pilot and manufacturing data to senior leadership, enabling key technical and program decisions.
- Successfully scaled fermentation and CFPS processes from ambr 250 systems to 30 L and 4 kL.
- Reviewed and approved CDMO quality documents (deviations, change controls, sampling plans, satellite plans).
- Performed statistical process control analysis, identifying more than \$75K per batch in potential operational improvements.
- Collaborated closely with the process development team to stabilize process performance.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Process Engineer | GOOD Meat (Eat Just), Alameda, CA

Jun 2022 - Mar 2024

- Led FAT and SAT execution for new stainless-steel and single-use cell culture equipment supporting pilot- and manufacturing-scale cultivated meat cell culture production.
- Led process scale-up and optimization studies involving kLa, P/V, tip speed, and parameter screening.
- Managed implementation of a plant-wide Ignition SCADA and historian integrating Siemens and Allen-Bradley PLC systems.
- Authored SOPs and batch records for media preparation, bioreactor, and harvest operations.
- Defined scale-up parameters from flask culture to a 3.5 kL bioreactor, successfully achieving at-scale KPI targets.
- Built an Excel visualization tool that reduced routine graph-generation time by 95%.
- Trained operations personnel on new equipment operation and maintenance.
- Designed the sitewide CIP strategy and authored and executed cleaning validation protocols.
- Achieved a 100% batch success rate in 2023 at a new animal cell culture facility and implemented low-capital modifications that cut operating costs by more than 50%.

Manufacturing Process Engineer | Codexis, Redwood City, CA

Jun 2019 - May 2022

- Doubled manufacturing capacity for an E. coli-based 15 L protein production line by repairing critical equipment and qualifying additional bioreactors.
- Managed a \$3M equipment upgrade program replacing obsolete bioreactors, centrifuges, and lyophilizers.
- Served as SME for fermentation, filtration, centrifugation, homogenization, and lyophilization unit operations.
- Transferred bacterial fermentation processes into 15 L pilot bioreactors and scaled them to 650 L stainless-steel systems.
- Implemented a 21 CFR Part 11-compliant CMMS for asset, calibration, and maintenance records.
- Managed the reliability program to achieve 100% on-time calibration and maintenance.

Manufacturing Engineer | AstraZeneca, Redwood City, CA

Aug 2016 - Jun 2019

- Commissioned and qualified a new API production line and reduced campaign deviations by 93% through upgraded equipment, automation, leveled scheduling, hands-on training, and clear procedural controls.
- Implemented and validated a process that increased weekly production output by 60% with existing equipment and headcount.
- Increased batch success from 88% to 97% year over year through improved equipment reliability, simplified documentation, and robust training.
- Led the manufacturing continuous process validation initiative in alignment with FDA guidance.
- Reduced facility product changeover from three weeks to one through scheduling optimization and elimination of non-value-added steps.
- Reduced manual batch record entry by 50% through automated reporting and documentation improvements.
- Served as SME for homogenization, spray drying, and jet milling unit operations.

Process Engineer | Bayer, Berkeley, CA

Jul 2013 - Jul 2016

- Planned and executed analytical process equipment CAPEX projects valued up to \$1M, delivering within 5% of budget.
- Led a sitewide reliability initiative that replaced more than 100 obsolete instruments.
- Performed statistical analyses to establish process ranges for commercial products.
- Evaluated new sampling and measurement technologies for biopharmaceutical processes.
- Designed preventive maintenance and calibration plans.
- Trained more than 300 operators and maintenance technicians across the entire site on cell counters, pH and DO instruments, and metabolite analyzers.

EDUCATION & CREDENTIALS

B.S., Bioengineering & Materials Engineering: University of California, Berkeley | May 2013

Lean Six Sigma Green Belt: Institute of Industrial and Systems Engineers (IISE) | 2015

Google Data Analytics Certificate: Dec 2023